

Project Design Phase

Problem – Solution Fit Template

Date	29 September 2026
Team ID	SWTID-2026-8776
Project Name	AI WeatherWise API
Maximum Marks	2 Marks

Chevalier T. Thomas Elizabeth College for Women

Team:

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Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who is your customer? I.e. working parents of 0-5 y.o. kids	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? I.e. spending power, budget, no cash, network connection, available devices.	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? I.e. pen and paper is an alternative to digital notetaking	Explore AS, differentiate
Focus on J&P, tap into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides.	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? I.e. customers have to do it because of the change in regulations.	7. BEHAVIOUR What does your customer do to address the problem and get the job done? I.e. directly related: find the right solar panel installer, calculate usage and benefits; Indirectly associated: customers spend free time on volunteering work (I.e. Greenpeace)	Focus on J&P, tap into BE, understand RC
Identify strong TR & EM	3. TRIGGERS What triggers customers to act? I.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.	10. YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? I.e. lost, insecure > confident, in control - use it in your communication strategy & design.		8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.	

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.

- ❑ **Understand the existing situation in order to improve it for your target group.**

Customer Problem	AI WeatherWise API Solution
Users find raw weather data difficult to understand and use for decision-making.	Gemini AI converts weather metrics into easy-to-understand weather summaries and actionable insights.
Users need personalized guidance for clothing, travel, and outdoor activities based on weather conditions.	AI-powered recommendations provide personalized suggestions according to current weather conditions.
Users frequently check weather for multiple cities and struggle to manage them efficiently.	Favorite Location Management allows users to save, view, update, and delete preferred locations.
Users need secure access to their weather preferences across sessions.	JWT Authentication and bcrypt password encryption ensure secure user accounts and protected access.
Users must switch between multiple applications to obtain weather data and recommendations.	AI WeatherWise combines real-time weather retrieval and AI-generated insights within a single platform.

Existing Situation

- Weather applications mainly provide numerical weather data.
- Users manually interpret temperature, humidity, and wind information.
- Personalized weather advice is limited.
- Managing multiple locations can be inconvenient.

Improved Situation with AI WeatherWise API

- Real-time weather information is presented in a user-friendly manner.
- AI-generated summaries simplify weather interpretation.
- Personalized recommendations improve decision-making.
- Favorite locations are managed from a single secure account.
- Users receive weather data and AI insights through one integrated platform.